
Month 8 Review and Recovery Plan



ARES
CONSORTIUM



Solis



THARSIS



Meridian



helix

Project Status

Four things have gone wrong at month 8. None is a failure of effort. All four follow from one document nobody has written: an interface contract saying what crosses between the four partners' models, in what units, on which clock, owned by whom.

Solis and Tharsis have yet to exchange anything on time.

Solis sizes generating capacity over multi-year horizons, Tharsis simulates habitat demand hour by hour, and their one exchange is two months late and in a format the receiver did not plan for. Nobody wrote down when it was due or what it should contain.

Tharsis and Helix disagree about what the shared variables mean and who owns them.

Helix is building the translation layer between the partners' models. With no owner named for those variables, it read them its own way and built its tooling on that reading.

Meridian says the interface cannot carry a safety claim.

Meridian is the risk laboratory that sets what counts as acceptable service and judges the evidence. It needs every exchanged number to arrive with a stated uncertainty, and none of them do.

Nobody has shown the two models have to run together.

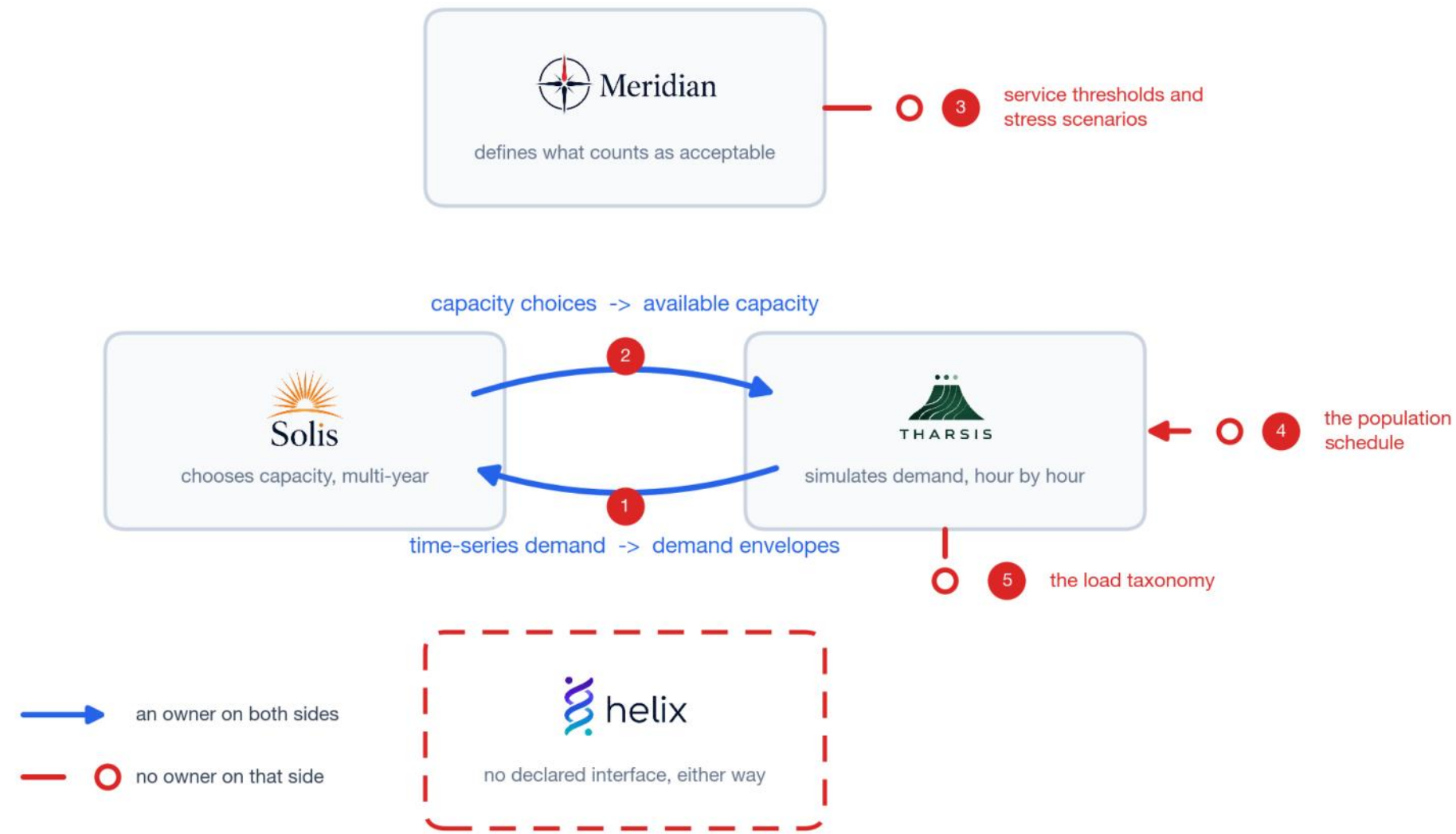
The conversions between Tharsis's hourly demand and Solis's multi-year capacity plan belong to no partner, so the coupling has never been tested.

Second-year funding depends on one end-to-end demonstration, and there is no demonstration.

Ownership Gap

Solis sizes generation over years, Tharsis simulates demand hour by hour, Meridian sets what counts as acceptable. Five things their models need that no partner produces. Of six possible partner pairs, only two state any exchange at all.

- 1. Hourly demand into a planning bound.
Nobody turns Tharsis's hour-by-hour demand into the bound Solis plans against.
- 2. Installed capacity into available.
What is really there after dust and outages. Tharsis invents its own number.
- 3. Service thresholds and stress scenarios.
Meridian writes both and sends them to nobody.
- 4. The population schedule.
Who is on Mars doing what. It drives every demand profile and no partner produces it.
- 5. Which loads can be shed.
In a storm, some load can wait and some cannot. Undeclared.



Model Fidelity

Tight coupling between the models is an absolute requirement. The question is at what fidelity this model becomes admissible evidence for a design decision that kills people if it is wrong.

A dust storm collapses generation while the load that keeps people alive stays flat.

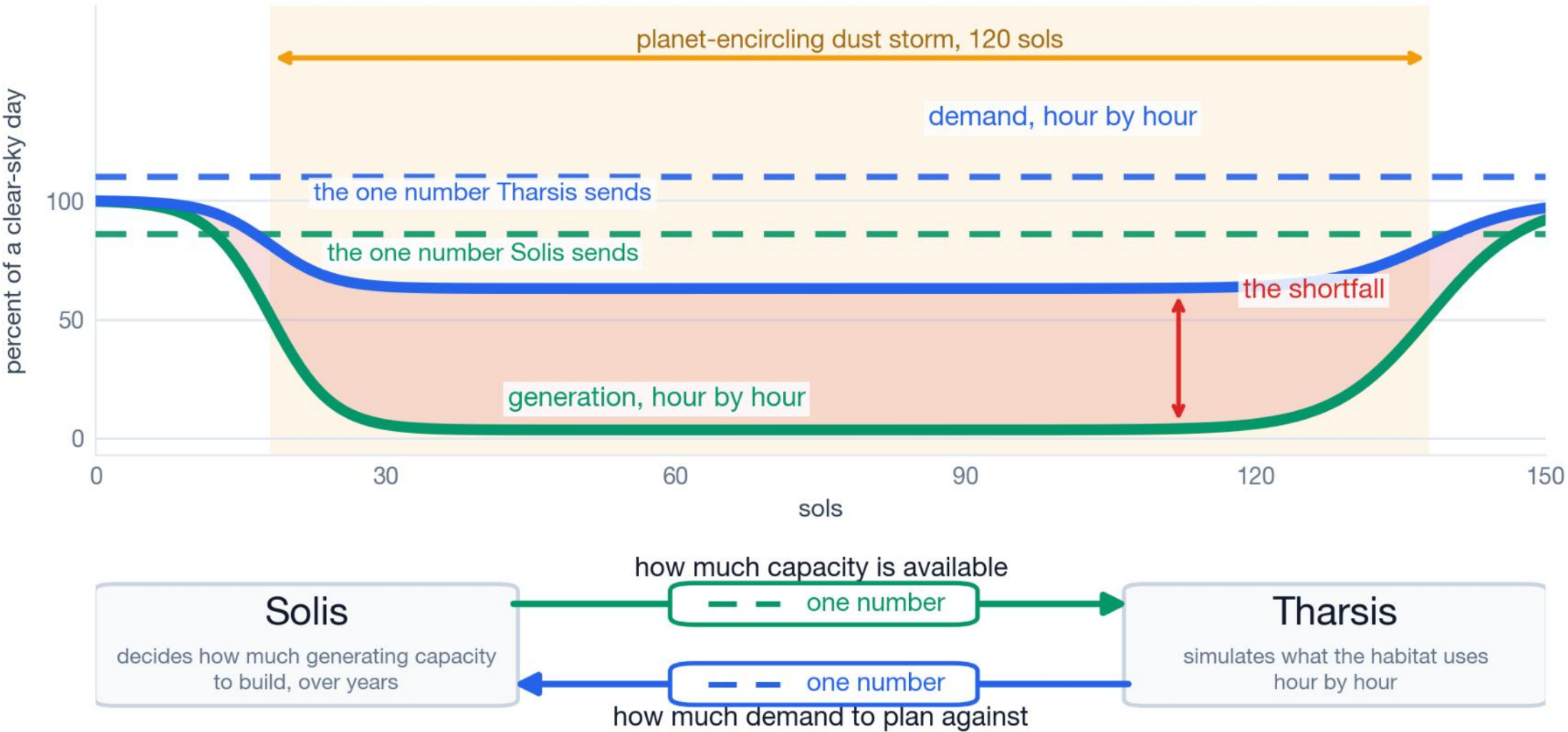
NASA's own baseline holds habitat power constant through a storm and sheds rover and drill to zero.

The interface sends one bound, so the two effects arrive separately.

Each model sees a worst case for its own side, never both at once.

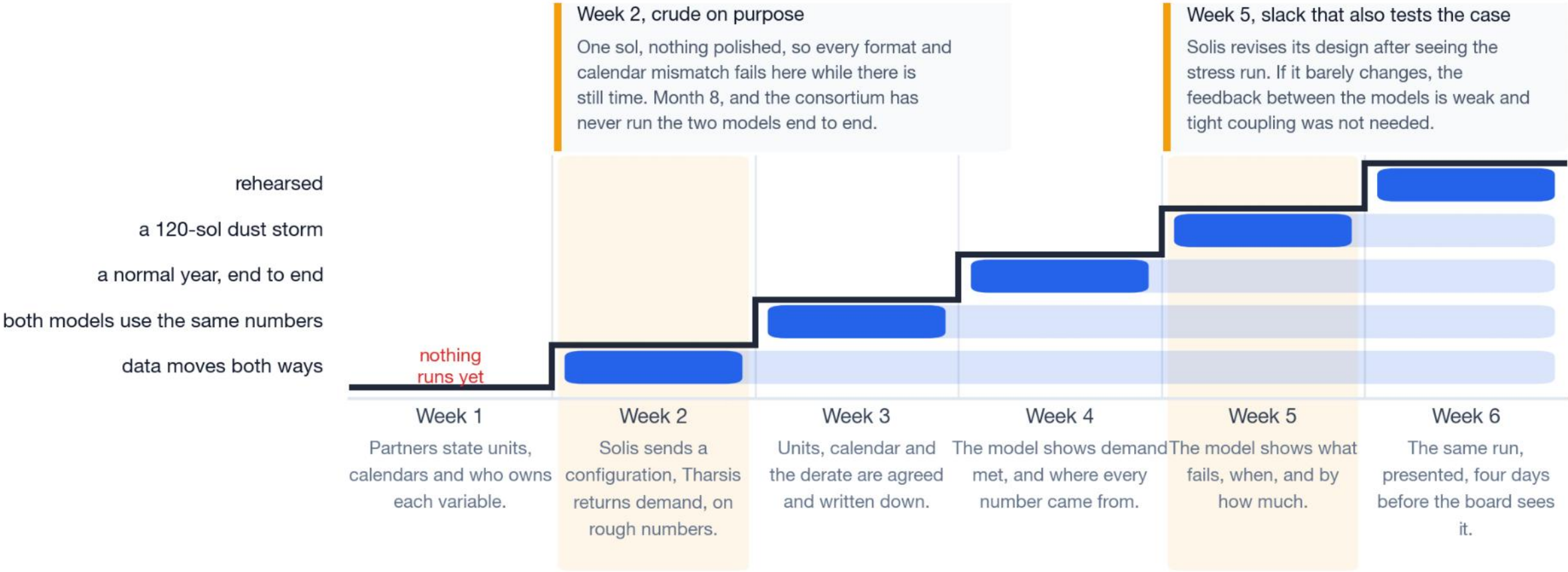
Somebody signs a habitat design on the strength of this model.

Below some level of detail it is not weak evidence for that decision, it is no evidence, and Meridian, the risk laboratory, has already said the interface cannot support the risk claims made from it.



Delivery Approach

From week 2 the demonstration runs end to end, Solis's capacity plan into Tharsis's hourly simulation and back. Every week after that adds to a run that already works.



Week 2 runs end to end on crude inputs.
One sol, nothing polished. Every format mismatch fails while there is time to fix it.

Month 8, and it has never run end to end.
Nothing has been tested between the two models.

Nothing is taken apart to add the next piece.
Worst case in week 6 is week 5's version.

Week 5 also tests this recommendation.
Solis revises its design after seeing the stress run. If it barely changes, the feedback between the models is weak and they did not need to run together.

Six Week Plan

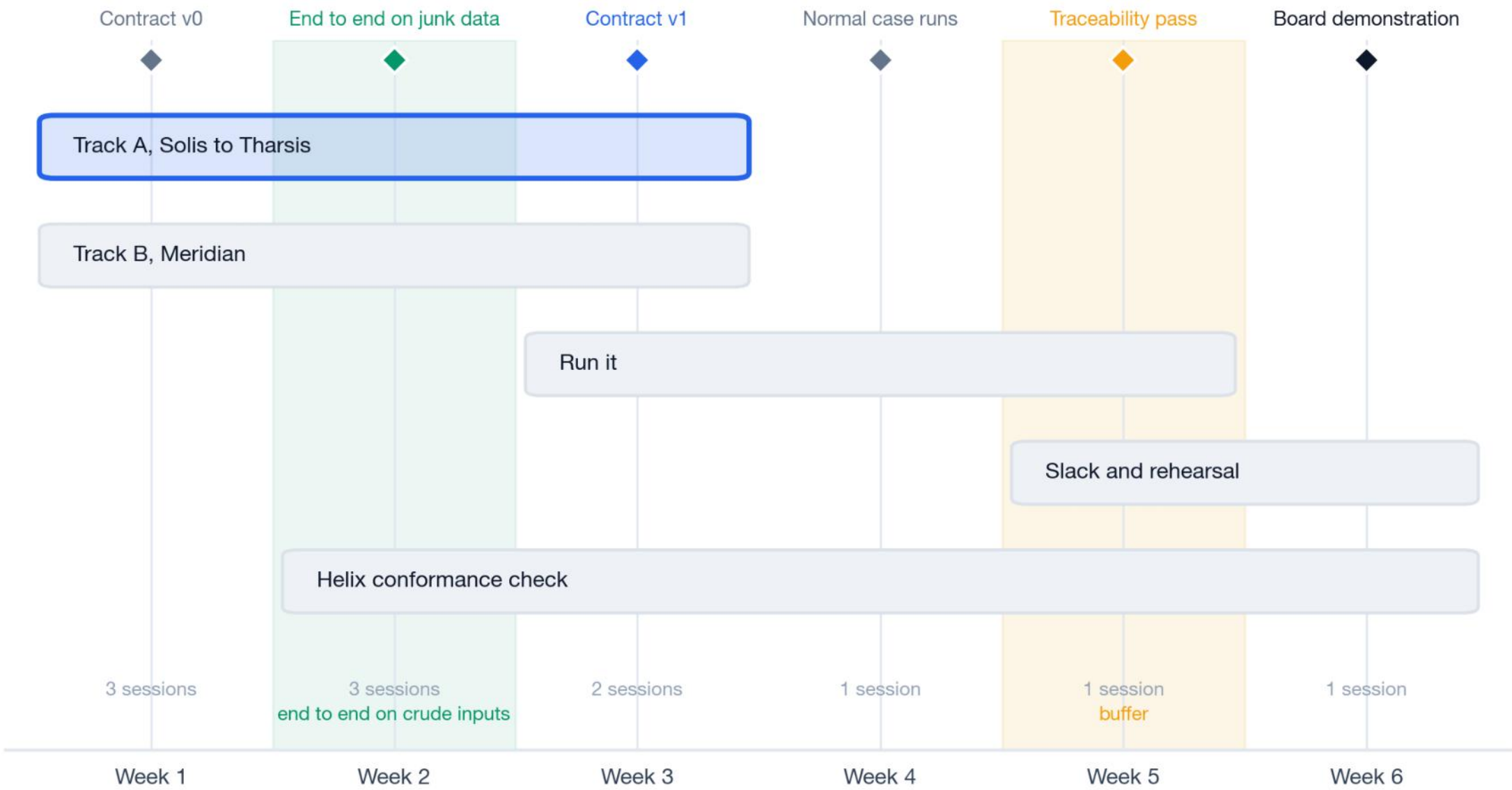
The exchange between Solis's multi-year capacity plan and Tharsis's hourly simulation is the critical path. Meridian's requirements run alongside it. Week 5 absorbs a slip and week 6 rehearses.

Track A, Solis to Tharsis.
Their one exchange is the one that already failed, and nothing else runs until it works.

Track B, Meridian.
From week 1, because the risk laboratory sets what the contract has to be able to carry.

Weeks 3 and 4.
The normal case first, then the stress event, with the two models iterating.

Week 6.
Every number traced back to the model, contract, schedule version and threshold set that produced it. That is the deliverable.



Meeting Plan

Every decision in the plan is made in the first three weeks. Weeks 4 to 6 review work that has already been agreed.

Front loaded.
Six of ten sessions fall in weeks 1 and 2.

Decisions early.
Five sessions settle something, all by week 3.

Helix hears, never decides.
In seven rooms, deciding in none.

Light back half.
One review a week, plus a short standup.

WEEK 1 Contract v0	WEEK 2 End to end on junk data	WEEK 3 Contract v1	WEEK 4 Normal case runs	WEEK 5 Traceability pass	WEEK 6 Board demonstration
Kickoff The diagnosis. The contract is the object. S T M H 90 min	Derate session How capacity degrades. Needs Meridian. S T M H 90 min	Stress case selection One stress event, named, with a duration. S T M H 60 min	Normal case review The normal case, end to end. S T M H 90 min	Stress case review The stress event, correlation preserved. S T M H 90 min	Dress rehearsal Rehearsed four days before the board. no attendee list half day
Interface working session Calendar, epoch, units, encoding. S T M H half day	Contract v1 sign-off Every cell filled, or visibly empty. S T M H half day	Integration standup Blockers only. S T M H 30 min weekly			
Evidence elicitation What makes a resilience claim credible. S T M H 90 min	Walking skeleton review What broke when it ran end to end. S T M H 60 min				
S in the room S holds the ruling S not in the room S Solis T Tharsis M Meridian H Helix					

Partner Alignment

No partner needs a new incentive to sign an interface contract. Each is already paying for the absence of one. Two of those costs are on the month 8 record, two are argued, and the difference matters.

Helix Dynamics, building the schema translation layer between the models, is the clearest case.

It has already built tooling against a provisional reading of interface variables whose meaning and ownership it and Tharsis still dispute. If that reading is wrong, the tooling is rework.

Meridian Risk Laboratory, which defines failure conditions and acceptable service levels, has already asked for this.

It states that the interface cannot represent failure or uncertainty well enough to support credible mission-risk claims. Its own product, the risk judgement, is what the gap blocks.

Solis Systems, sizing generation and storage capacity over multi-year horizons, is exposed on the record.

Its one exchange to Tharsis is two months late and the earliest output is not in the format the receiver planned around. Nothing on file says what that output owed or when, so it can miss the same way twice.

Tharsis Habitat Institute, the university team simulating habitat demand hour by hour, is the receiver.

Its inputs are capacity and operating constraints another partner produces, so the delay is time it cannot recover by working harder, and it planned around a format that never arrived.

The first two are stated in the record. The last two are argued, from the late exchange and from the scale mismatch nobody has yet shown a coupling can bridge. That mismatch is what Solis and Tharsis share, and the contract settles what has to cross at all.

Decisions Required

Four decisions. Three change only what the coordinator may ask for. The fourth narrows what week six delivers.

Endorse the diagnosis.

Five things the models need that no partner produces, all on the boundary between Solis's capacity plan and Tharsis's hourly simulation.

Back the sequence.

Solis to Tharsis first, Meridian's evidence requirements alongside. Nothing else starts until those two models exchange successfully.

Give the coordinator standing to make each partner write its side down.

What crosses each boundary, in what units, on which clock, and who owns it. Methods and code stay with the partner.

Accept a narrow demonstration.

One settlement configuration, one stress event, fully traceable. Not a platform, and not four couplings. This is the decision that costs you something.

In return, at week six.

That configuration under a normal case and a material stress event, every number traceable to the model, contract, schedule version and threshold set that produced it. And a test in week five that could tell you this recommendation was wrong.